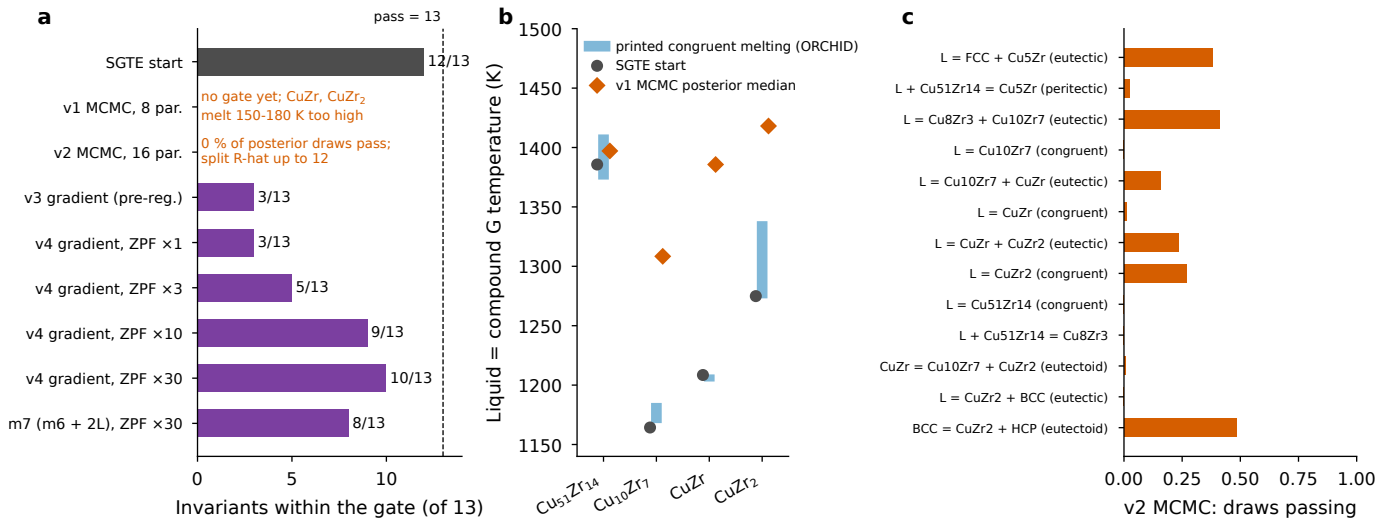
a
pass = 13
SGTE start
12/13
v1 MCMC, 8 par.
no gate yet; CuZr, CuZr2
melt 150-180 K too high
v2 MCMC, 16 par.
0 % of posterior draws pass;
split R-hat up to 12
v3 gradient (pre-reg.)
3/13
v4 gradient, ZPF ×1
3/13
v4 gradient, ZPF ×3
5/13
v4 gradient, ZPF ×10
9/13
v4 gradient, ZPF ×30
10/13
m7 (m6 + 2L), ZPF ×30
8/13
0
5
10
Invariants within the gate (of 13)
b
printed congruent melting (ORCHID)
SGTE start
v1 MCMC posterior median
Liquid = compound G temperature (K)
1500
1450
1400
1350
1300
1250
1200
1150
Cu51Zr14
Cu10Zr7
CuZr
CuZr2
c
L = FCC + Cu5Zr (eutectic)
L + Cu51Zr14 = Cu5Zr (peritectic)
L = Cu8Zr3 + Cu10Zr7 (eutectic)
L = Cu10Zr7 (congruent)
L = Cu10Zr7 + CuZr (eutectic)
L = CuZr (congruent)
L = CuZr + CuZr2 (eutectic)
L = CuZr2 (congruent)
L = Cu51Zr14 (congruent)
L + Cu51Zr14 = Cu8Zr3
CuZr = Cu10Zr7 + CuZr2 (eutectoid)
L = CuZr2 + BCC (eutectic)
BCC = CuZr2 + HCP (eutectoid)
0.00
0.25
0.50
0.75
1.00
v2 MCMC: draws passing